

PUBUDU GUNASEKARA

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MS Computer Science (Northeastern) • Seeking Software Engineering Co-op: January - August 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences - San Jose, CA Jan 2026 - Present

M.S. in Computer Science | GPA: 4.0 / 4.0 | Expected 2028

Conestoga College - Waterloo, ON, Canada 2022 - 2023

Postgraduate Certificate, Mobile Solutions Development | GPA: 3.74 / 4.0

Victoria University - Melbourne, Australia 2018 - 2021

Bachelor of Information Technology (Distinction)

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL, C++

Backend & Distributed: Spring Boot, Node.js / Express, Apache Kafka, Redis, PostgreSQL, MongoDB, REST APIs, microservices, hexagonal architecture

Infra & DevOps: Docker, Docker Compose, GitHub Actions, Jenkins, CI/CD, Prometheus, Grafana, AWS

Frontend: React, Next.js, React Native, Vite, Redux

EXPERIENCE

Gunasekara Transport - Web Developer Dec 2025 - Jun 2026 • Remote

- Built the corporate website and web app interfaces, plus an internal business management system (HR, operational workflows, and reporting) that replaced manual processes, using React.js, JavaScript, PHP, WordPress, and Docker.
- Managed web infrastructure including domain configuration and email gateway administration.

Virtusa - Associate Engineer, QA Jun 2021 - Mar 2022 • Colombo, Sri Lanka

- Built Java and Selenium WebDriver test automation for a British Telecommunications platform, designing and executing test cases and test plans to verify system functionality across agile sprints.
- Led test planning and ran client demos and daily stand-ups for the module under the QA lead's delegation, and collaborated with developers to validate bug fixes.
- Tracked QA metrics including defect densities and open defect counts, working across CST Environment, Camunda Cockpit, RabbitMQ, Swagger UI, and MQ Console.

PROJECTS

Distributed Task Scheduler

Java 21, Spring Boot 3.5, Apache Kafka, Redis, PostgreSQL, Prometheus, Grafana, Docker, GitHub Actions • [Code](#)

- Built a distributed task scheduler on a hexagonal architecture enforced at build time via ArchUnit, with at-least-once delivery over Kafka and a state-machine guard that prevents duplicate execution on redelivery.
- Implemented a Redis distributed lock (atomic SETNX acquire, Lua compare-and-delete release) and an exponential-backoff retry scheduler (10s, 30s, 90s) with dead-letter handling.
- Instrumented the system end-to-end with Prometheus metrics and Grafana dashboards; containerized with Docker Compose and automated build/test in GitHub Actions CI.

AI Code Review Assistant

React 19 / Vite, Node.js / Express, PostgreSQL, Redis, gpt-4o-mini, Docker, AWS (EC2 + Amplify) • [Code](#) • [Live Demo](#)

- Built and deployed a full-stack app that reviews a selected GitHub pull request via GitHub OAuth, returning severity- and category-tagged findings from gpt-4o-mini in a strict JSON schema.
- Engineered per-user Redis rate limiting with a fail-open policy and standard rate-limit headers, plus diff truncation to stay within the model's context budget.

Travel Day Scheduler

Python, Weighted A* Search • Graduate algorithms project (team) • [Code](#)

- Implemented a weighted / anytime A* itinerary planner with iterative weight escalation and heap pruning to bound memory under large search spaces; built the route-visualization frontend.

ACTIVITIES

- Northeastern University: Husky Code Club (Member) • Graduate Student Career Advisory Board (Member)